

Analyzing Computing Technology in the Wild

ITSC 2181 - Module 1 Unit 1: Computer Architecture

Introduction

This assignment will have you reflect on multiple pieces of technology, analyzing whether or not they are computing systems, what types of computational tasks they may undertake, and beginning to identify the technology involved in doing so. You will do this by first analyzing your computer for the assignment, then a piece of technology you own, then a piece of technology you find online, and then a piece of assistive technology.

For each of the devices, you will answer the questions and prompts included below them, with the goal being identifying the nature of the device you are analyzing.

Device 1: Your Computer

For your first device, analyze a computer that you own. This does not have to be the computer you are completing this assignment on.

- Name of device: Acer Swift 3
- Link to device: <https://www.acer.com/us-en/laptops/swift/swift-3-intel/pdp/NX.A5UAA.006>
- Is this device a computer?: Yes, it is a laptop.
- What kind of tasks is this device for?: This laptop is generally for school use. It can handle everyday tasks such as running programs, browsing the internet, etc.
- Do you think this device has expensive hardware (>\$500)?: No, it is around \$399 and is considered budget-tier.
- How much does this device cost?: The current model is discontinued, but refurbished models retail about \$399.

Device 2: Another Device You Own

For your second device, analyze a device that you own that is not your computer. This could be a phone, smart device, or something else.

- Name of device: OnePlus Open

- Link to device: <https://www.oneplus.com/us/open>
- Is this device a computer?: No, it is a mobile device.
- What kind of tasks is this device for?: The mobile phone allows portable communication, access to the internet, running programs, etc.
- Do you think this device has expensive hardware (>\$500)? Yes, it's a product that features relatively new technology.
- How much does this device cost?: The device retails for about \$800.

Device 3: A Device From the Internet

For your next device, find a device from the internet to analyze; this could be anything from any website, but make it unique from your previous two devices.

- Name of device: Yamaha MX61BK Keyboard Synthesizer
- Link to device: <https://www.americanmusical.com/p/yamaha-mx61-v2-61-key-keyboard-synthesizer?sku=YAM-MX61BK>
- Is this device a computer?: No, this device is not a computer
- What kind of tasks is this device for?: This device is an electronic musical instrument that generates and modifies audio signals.
- Do you think this device has expensive hardware (>\$500)? Yes, this hardware should be relatively retro but around \$500.
- How much does this device cost?: This device retails for about \$899

Device 4: Assistive Technology Example

For this final device, you'll be investigating a specific device from the internet. You'll likely have to perform additional research to find information on this device, especially for the later questions. Here is a [listing for the accessible technology device you will analyze](#).

- Name of device: Lightwriter SL50
- Link to device: <https://www.abilia.com/en/our-products/communication/lightwriter-sl50>
- Is this device a computer?: No, it would not be considered a computer, as it is missing certain key computer features.
- What kind of tasks is this device for?: This device is used for text-to-speech communication to assist with speech impairment.

Analyzing Computing Technology in the Wild

- Do you think this device has expensive hardware (>\$500)? Yes, as this device is specialized medical technology, it typically costs more than everyday electronics.
- How much does this device cost?: This device retails for about \$7,795.

Reflections

Now that you have analyzed four devices using this framework, answer the following questions in a few sentences each.

- Which device(s) are you most familiar with, and which device(s) are you least familiar with?
I am most familiar with everyday electronics such as the laptop and mobile phone. A device I am least familiar with is the Lightwriter SL50. Unlike common electronics, I have never used such specialized equipment.
- Which device(s) are most important for day-to-day life, and which device(s) are the least important?
I believe it depends on the user. As I am a college student, a laptop and phone are essential. Similarly, while a Lightwriter is unimportant to me, it may be a critical item for someone with Parkinsons.
- Which device is the most expensive? Why do you think that is? The Lightwrite SL50 is the most expensive item, and there are multiple factors. First, it's expensive hardware for that lightweight efficiency. Second, it's a specialized device, meaning low production volume. Third, it's a medical device, meaning strict regulation affecting final cost.

Submission Instructions

To submit this assignment, download this document either as a PDF or DOCX file, and submit it on Canvas.